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Mathematics for New Technologies in Finance, Mini-project

Part I

1. Θ_0 is the subset of all admissible trading strategies ($\theta \in \Theta$) such that the trading strategy weights (values in $\mathbb{R}^d$) at time t can be represented by the gradient of some $\varphi \in \mathcal{D}$ at X_t, i.e. $\theta_t = \nabla\varphi(X_t)$.

2. Let X_t ($t \geq 0$) be a d-dimensional OU process with drift matrix $\beta \in \mathbb{R}^{d\times d}$ and instantaneous covariance matrix $c \in \mathbb{R}^{d\times d}$, defined by $dX_t = -\beta X_t dt + \sqrt{c}\, dW_t$. The wealth process is $V_T^\theta = \mathcal{E}\left(\int_0^T \theta_t^\top dX_t\right)$, which implies $dV_t^\theta = V_t^\theta \theta_t^\top dX_t$. Applying Itô's lemma to $f(v) = \log(v)$:

$$\log(V_T^\theta) = \log(V_0^\theta) + \int_0^T \frac{1}{V_s^\theta}\, dV_s^\theta - \frac{1}{2}\int_0^T \frac{1}{(V_s^\theta)^2}\, d\langle V^\theta\rangle_s$$

Since $d\langle X\rangle_t = c\, dt$ and $d\langle V\rangle_t = (V_t^\theta)^2 \theta_t^\top c\, \theta_t\, dt$, the expression becomes:

$$\log(V_T^\theta) = \log(V_0^\theta) + \int_0^T \theta_s^\top dX_s - \frac{1}{2}\int_0^T \theta_s^\top c\, \theta_s\, ds$$

For $\theta \in \Theta_0$, we substitute $\theta_s = \nabla\varphi(X_s)$:

$$\log(V_T^\theta) = \log(V_0^\theta) + \int_0^T \nabla\varphi(X_s)^\top dX_s - \frac{1}{2}\int_0^T \nabla\varphi(X_s)^\top c\, \nabla\varphi(X_s)\, ds$$

Applying the multivariate Itô's lemma to $\varphi(X_T) \in C^2(E)$, we obtain:

$$\varphi(X_T) = \varphi(X_0) + \int_0^T \nabla\varphi(X_s)^\top dX_s + \int_0^T \left(\frac{1}{2}\operatorname{Tr}(c\, \nabla^2\varphi(X_s))\right) ds$$

Substituting this back yields:

$$\log(V_T^{\theta_0}) = \log(V_0^{\theta_0}) + \varphi(X_T) - \varphi(X_0) - \frac{1}{2}\int_0^T \left(\operatorname{Tr}(c\, \nabla^2\varphi(X_s)) + \nabla\varphi(X_s)^\top c\, \nabla\varphi(X_s)\right) ds$$

The growth rate is defined as $g(\theta; P) = \sup\left\{\gamma \in \mathbb{R} : \liminf_{T\to\infty} \frac{1}{T}\log(V_T^\theta) \geq \gamma\right\}$. We evaluate the limit:

$$\liminf_{T\to\infty} \frac{1}{T}\log(V_T^{\theta_0}) = \liminf_{T\to\infty} \frac{1}{T}\left(\log(V_0^{\theta_0}) + \varphi(X_T) - \varphi(X_0) - \frac{1}{2}\int_0^T \dots\, ds\right)$$

The boundary terms $\frac{1}{T}\left(\log(V_0^{\theta_0}) + \varphi(X_T) - \varphi(X_0)\right) \to 0$ as $\varphi \in \mathcal{D}$ and since we assume that the market dynamics P are ergodic with a stationary density p, so that X satisfies the tightness property. By the ergodic theorem, the remaining integral becomes, with $p(x)$ the stationary distribution of X:

$$\liminf_{T\to\infty} \frac{1}{T}\left(-\frac{1}{2}\int_0^T \dots\, ds\right) = -\frac{1}{2}\int_E \left(\operatorname{Tr}(c\, \nabla^2\varphi(x)) + \nabla\varphi(x)^\top c\, \nabla\varphi(x)\right) p(x)\, dx$$

This is a constant, so we conclude that $g(\theta; P) = -\frac{1}{2}\int_E \left(\operatorname{Tr}(c\, \nabla^2\varphi(x)) + \nabla\varphi(x)^\top c\, \nabla\varphi(x)\right) p(x)\, dx$ by setting γ equal to this value. Hence, it is independent of the measure P, so Θ_0 has the growth rate invariance property over Π. This means that when we are maximising over Θ_0, the adversary's actions and choice of measure P do not impact our asymptotic growth rate, so the simplified problem is now $\sup_{\theta\in\Theta_0} g(\theta)$.

3. Consider $\int_E \frac{\operatorname{Tr}(A(x)\nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} dx$. Since the trace is a linear operator, we can rewrite it as: $\int_E \sum_{i,j}^d e^{-\varphi(x)} A_{ij}(x)\partial_i\partial_j(e^{\varphi(x)})dx$ Consider $\int_E e^{-\varphi(x)} A_{ij}(x)\partial_i\partial_j(e^{\varphi(x)})dx$. Applying IBP let $\partial_i v = \partial_i\partial_j e^{\varphi(x)}$ and $u = e^{-\varphi(x)} A_{ij}(x)$. This gives $v = \partial_j e^{\varphi(x)}$ and $\partial_i u = (-\partial_i\varphi(x)A_{ij}(x) + \partial_i A_{ij}(x))\, e^{-\varphi(x)}$. So the integral evaluates to:

$$\left[\partial_j e^{\varphi(x)} e^{-\varphi(x)} A_{ij}(x)\right]_E - \int_E \partial_j e^{\varphi(x)} \partial_i u dx = \left[\partial_j\varphi(x) e^{\varphi(x)} e^{-\varphi(x)} A_{ij}(x)\right]_E - \int_E \partial_j e^{\varphi(x)} \partial_i u dx$$

which becomes $0 - \int_E \partial_j e^{\varphi(x)}\partial_i u dx$ as $\partial_j\varphi(x) = 0$ on bounds of E as φ is compactly supported. Consider the remaining integral $\int_E \partial_j e^{\varphi(x)}\partial_i u dx$:

$$\int_E \partial_j\varphi(x) e^{\varphi(x)} e^{-\varphi(x)} \left(-\partial_i\varphi(x) A_{ij}(x) + \partial_i A_{ij}(x)\right) dx = \int_E \left(-\partial_j\varphi(x)\partial_i\varphi(x) A_{ij}(x) + \partial_j\varphi(x)\partial_i A_{ij}(x)\right) dx$$

So the expression becomes:

$$\int_E \sum_{i,j}^{d} \partial_j \varphi(x) \left(\partial_i \varphi(x) A_{ij}(x) - \partial_i A_{ij}(x)\right) dx = \int_E \left(\nabla\varphi(x)^\top A(x) \nabla\varphi(x) dx - \nabla\varphi(x)^\top \mathrm{div} A(x)\right)$$

We observe that this resulting expression has no Hessian term. By expanding $\nabla^2 e^{\varphi}$ via the chain rule, we see that the term in question 3 is equal to $-2g$ from 2. We have therefore proved that g can be rewritten as an expression that has no Hessian term, simplifying the problem.

4. Recall $\mathrm{div}A = \sum_{i=1}^d \partial_i A_i$. Here $\mathrm{div}A = \mathrm{div}(cp) = \mathrm{div}(pc)$, where $p = p(x)$ is the scalar density. This expands to $\mathrm{div}(pc) = \nabla p \cdot c + p \nabla \cdot c$. Since c is a constant matrix, $\nabla \cdot c = 0$, so $\mathrm{div}A = \nabla p \cdot c$. Given $A^{-1} = (cp)^{-1} = \frac{1}{p} c^{-1}$, it follows that $A^{-1}\mathrm{div}A = \frac{1}{p} c^{-1}(\nabla p \cdot c)$. Since $c^{-1}c = I$, this simplifies to $A^{-1}\mathrm{div}A = \frac{\nabla p}{p} = \nabla \log p$. This result confirms that our model satisfies the 'Gradient Case' condition in Section 7.1 of Itkin et al. (2022)

5. Using the result from question 3, $A(x) = p(x)c$ and the identity from question 4, $A^{-1}\mathrm{div}A = \nabla \log p(x)$, we substitute $\mathrm{div}A = A\nabla \log p(x)$:

$$g(\varphi) = \frac{1}{2} \int_E \left(\nabla\varphi(x)^\top c \nabla \log p(x) - \nabla\varphi(x)^\top c \nabla\varphi(x)\right) p(x) dx$$

Now we compute $\nabla \log p(x)$. With $p(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp(-\frac{1}{2} x^\top \Sigma^{-1} x)$, we have $\log p(x) = \log(C) + (-\frac{1}{2} x^\top \Sigma^{-1} x)$. Thus $\nabla \log p(x) = -\Sigma^{-1} x$. We substitute this in to get:

$$g(\varphi) = \frac{1}{2} \int_E \left(\nabla\varphi(x)^\top c(-\Sigma^{-1}x) - \nabla\varphi(x)^\top c \nabla\varphi(x)\right) p(x) dx$$

We can see that the problem is Linear Quadratic (LQ) as the objective is linear in the control variable $\nabla\varphi$ through the first term and quadratic in $\nabla\varphi$ through the second term, while $\nabla \log p(x) = -\Sigma^{-1}x$ is linear in x. The problem is now $\sup_{\theta \in \Theta_0} g(\theta, P)$, which is equivalent to $\sup_{\varphi \in \mathcal{D}}$ since $\theta = \nabla\varphi$. Hence, we differentiate the integrand with respect to $\nabla\varphi$. To solve the optimization problem, we set the derivative to zero: $-c\Sigma^{-1}x - 2c\nabla\varphi(x) = 0$. Since c is a symmetric positive definite matrix ($c \in S^d_{++}$), it is non-singular and thus invertible. We have $2\nabla\varphi(x) = -\Sigma^{-1}x \implies \nabla\varphi(x) = -\frac{1}{2}\Sigma^{-1}x$. Integrating this gradient gives the quadratic form $\varphi(x) = -\frac{1}{4} x^\top \Sigma^{-1} x + K$.

6. The robust-optimal trading strategy is given by $\theta(x) = \nabla\varphi(x) = -\frac{1}{2}\Sigma^{-1}x$. This strategy is "robust" because in 2 we showed that the growth rate is invariant to the measure P, and it is "optimal" because it is the maximum of the growth rate for a given stationary density $p(x)$. We see that the portfolio weights are proportional to $-x$ indicating a mean-reverting strategy. As x increases, the optimal strategy takes a shorter position, trading against the asset prices x.

7. To compute the robust-optimal growth rate g^*, we substitute $\nabla\varphi^*(x) = -\frac{1}{2}\Sigma^{-1}x$ into the simplified expression:

$$g^* = \frac{1}{2} \int_E \left(\nabla\varphi^*(x)^\top c \nabla \log p(x) - \nabla\varphi^*(x)^\top c \nabla\varphi^*(x)\right) p(x) dx.$$

Since $\nabla \log p(x) = -\Sigma^{-1}x$, this becomes

$$g^* = \frac{1}{2} \int_E \left(\frac{1}{2} x^\top \Sigma^{-1} c \Sigma^{-1} x - \frac{1}{4} x^\top \Sigma^{-1} c \Sigma^{-1} x\right) p(x) dx = \frac{1}{8} \int_E x^\top \Sigma^{-1} c \Sigma^{-1} x \, p(x) dx.$$

Since $X \sim N(0, \Sigma)$ $\mathbb{E}\left[X^\top \Sigma^{-1} c \Sigma^{-1} X\right] = \mathrm{Tr}\left(\Sigma^{-1} c \Sigma^{-1} \Sigma\right) = \mathrm{Tr}(c\Sigma^{-1})$ Hence, $g^* = \frac{1}{8}\mathrm{Tr}(c\Sigma^{-1})$. We observe that the robust-optimal growth rate depends only on the instantaneous covariance c and the invariant covariance Σ. In particular it does not depend on $P \in \Pi$, which is consistent with the growth-rate invariance property.

Part II

1. The general form of the d-dimensional Ornstein-Uhlenbeck process is given by $dX_t = B(\eta - X_t)dt + \sigma dW_t$ where $X_t \in \mathbb{R}^d$ is the state vector, $B \in \mathbb{R}^{d\times d}$ is the drift matrix, $\eta \in \mathbb{R}^d$ is the long-term mean vector, and $\sigma \in \mathbb{R}^{d\times d}$ is the diffusion matrix. The adversary can only perturb the distribution under the constraint that the stationary distribution has density $p(x) = N(0, \Sigma)$, hence the long term mean η is fixed at 0. The instantaneous volatility is also constrained such that $\sigma\sigma^\top = c$, where c is a fixed matrix given in the problem formulation, so the adversary can perturb only the non-fixed parameter, the drift matrix B.

2. The probability density $p(x)$ of the stationary distribution satisfies the stationary Fokker-Planck equation, where the temporal derivative $\frac{\partial p}{\partial t}$ equals 0. Since the drift is $b(x) = -Bx$ and $\sigma\sigma^\top = c$, this is given by $0 = -\nabla \cdot ((-Bx)p(x)) + \frac{1}{2}\nabla \cdot (c\nabla p(x))$ Assuming that $p(x) = \mathcal{N}(0, \Sigma)$ and substituting $p(x) \propto \exp\left(-\frac{1}{2} x^\top \Sigma^{-1} x\right)$ we obtain the Lyapunov equation, which is the constraint that Nature must satisfy to maintain a valid stationary distribution.

$$\Sigma + \Sigma B^\top = \sigma\sigma^\top = c$$

3. The drift matrix B, has d^2 total parameters, but it must satisfy the constraint discussed above in 2. The equality above is symmetric i.e. it's an equality with a symmetric matrix on both sides, so there can only be $\frac{d(d+1)}{2}$ independent equations that must be satisfied. Hence there are $\frac{d(d+1)}{2}$ constrained DOF, so the Adversary has $d^2 - \frac{d(d+1)}{2} = \frac{d(d-1)}{2}$ degrees of freedom. Equivalently, the admissible drift matrices can be parameterized as $B = \frac{1}{2}c\Sigma^{-1} + Q\Sigma^{-1}, \qquad Q^\top = -Q$ where the skew-symmetric matrix Q has exactly $\frac{d(d-1)}{2}$ free parameters.
4. 5. We formulate the problem as minmax game between the agent, who tries to learn the best possible trading strategy given the worst case model and the adversary who tries to choose the worst possible drift for the OU model. This follows the GAN-style robust utility framework of Krach et al.

- **Agent ($\mathcal{N}_A$):** A feed-forward MLP with inputs current state X_t and normalized time t/T, and output the strategy, $\theta(X_t)$. It has two hidden layers with 64 neurons and LeakyReLU activations.
- **Adversary ($\mathcal{N}_N$):** A feed-forward MLP that outputs the parameters parametrizing the drift matrix, which has $\frac{d(d-1)}{2}$ degrees of freedom. Then we parameterize the admissible drift matrices directly by $B = \frac{1}{2}c\Sigma^{-1} + Q\Sigma^{-1}, \qquad Q^\top = -Q$. This ensures that the Fokker-Planck constraint $(B\Sigma + \Sigma B^\top = c)$ is satisfied. Since in this OU example with d = 2 the adversary has finitely many degrees of freedom, we use a direct trainable parameter vector for Nature rather than a neural network.

For training, we generate data synthetically using the Euler-Maruyama discretization of the Langevin dynamics as discussed in lectures. Following the framework proposed by Krach et al., we use alternating gradient descent-ascent. The loss function used is the discretized finite-horizon growth rate objective:

$$\widehat{g}_T = \frac{1}{T}\mathbb{E}\left[\sum_{k=0}^{N-1} \theta(X_k)^\top (X_{k+1} - X_k) - \frac{1}{2}\theta(X_k)^\top c\theta(X_k)\Delta t\right].$$

The Agent minimizes $-\widehat{g}_T$, while the Adversary minimizes $+\widehat{g}_T$. To aid with potential instability, what worked well was bounding both players' outputs. We also use a fixed random seed for reproducibility and evaluate the final learned strategy by freezing the Agent and checking its growth over a grid of admissible adversarial parameters q.
6. 7. In Figure 1, the left panel shows the learned growth rate during training, with the red dashed line corresponding to the theoretical value $g^* = 0.25$. The right panel shows the learned strategy field $\pi(X)$. The learned strategy is predominantly mean-reverting, with arrows pointing toward the origin. This is consistent with the analytical robust-optimal strategy $\theta^*(x) = -\frac{1}{2}\Sigma^{-1}x$ Since in our numerical experiment $\Sigma = 0.25I$, this becomes $\theta^*(x) = -2x$ The computed robust-optimal growth rate is 0.255030, reported as the worst-case learned growth after freezing the learned Agent and evaluating it over a grid of admissible parameters $q \in [-1, 1]$. This is very close to the theoretical value 0.25. Hence the learned strategy and growth rate are consistent with the analytical solution. This is consistent with the result from Itkin et al. that the robust-optimal strategy lies in the gradient class Θ_0. Although the neural network parameterizes the strategy directly, rather than enforcing the form $\theta = \nabla\varphi$, the learned field is close to the analytical gradient strategy $\theta^*(x) = -\frac{1}{2}\Sigma^{-1}x$, hence these results provide numerical evidence supporting the analytical result from Part I.

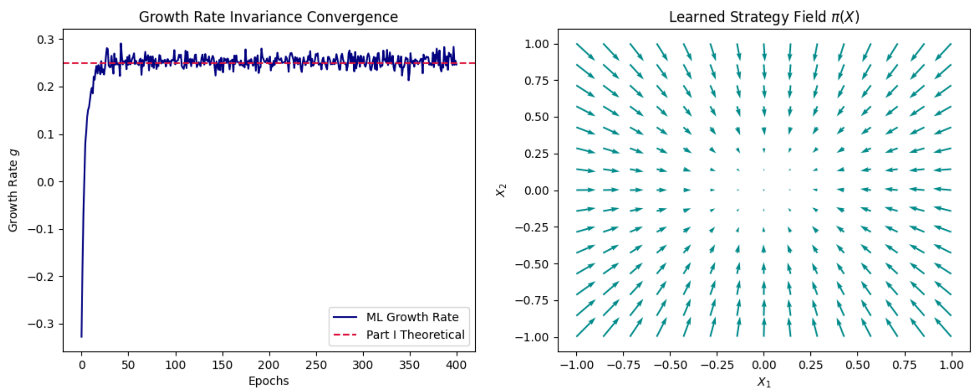

Figure 1: Training growth rate and learned robust-optimal strategy field.

Overall, this project confirms that the robust-optimal strategy for an OU market is a functionally generated portfolio, which is invariant to the drift of Nature, as suggested in Itkin et al.